

Saathveek Gowrishankar

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EDUCATION

University of Illinois Urbana-Champaign

Aug 2022 - May 2026

B.S. in Computer Engineering, Minor in Data Science

Study abroad at Nanyang Technological University, Singapore

Courses: Machine Learning, Tiny ML, Parallel Programming, Operating Systems, Software Engineering

EXPERIENCE

Machine Learning and Infra Intern

May 2025 - Aug 2025

Blue Cross and Blue Shield of Illinois

Chicago, IL

- Integrated machine learning into capacity planning, reporting to the Executive Director of Infrastructure.
- Developed a multi-model seasonal forecasting system to inform proactive CPU, RAM, and storage upgrades.
- Designed fallback strategies to simpler models and flagged insufficient data, maintaining forecast reliability.
- Optimized problem reporting for 2,700+ apps, improving speed 61% and coverage 37% to prevent server failure.
- Drove AIOps scalability planning by analyzing membership and distributed resource usage trends with PySpark.

Software Engineering Intern

May 2024 - Aug 2024

Mullbry, Inc. (Startup)

Rochester, NY

- Collaborated with the CEO to transform the borrower approval process into an efficient AI-assisted pipeline.
- Automated LLM inference via OpenAI API Python function calling, accelerating borrower approval times 30%.
- Integrated Plaid, credit, and self-reported data using Pandas to build accurate, equitable borrower profiles.
- Built a modular testing framework to automate LLM accuracy evaluations across diverse analytical scenarios.

LLM Linguistic Resource Researcher

Jan 2026 - Present

School of Information Sciences at UIUC

Champaign, IL

- Investigated data scarcity in LLM training for 80+ languages to identify challenges in equitable AI development.
- Simulated NLP filtering pipelines to reveal disproportionate data loss for low-resource African languages.
- Developed a sampling method to promote equitable representation of low-resource languages in NLP.

Human-Robot Interaction Researcher

Aug 2022 - Aug 2025

Human Factors and Aging Lab at UIUC

Champaign, IL

- Optimized data collection and automated citation generation in Python, cutting research timelines 40%.
- Built Lasso and Ridge regression models using scikit-learn and statsmodels to estimate trust in assistive robots.
- Presented research findings for two consecutive years at the Illinois Undergraduate Research Symposium.

PROJECTS

Custom Linux-style Operating System

Aug 2025 - Dec 2025

University of Illinois Urbana-Champaign

Champaign, IL

- Developed a RISC-V Linux-style OS that supports concurrency and safe resource sharing across kernel threads.
- Implemented a custom kernel-level file system backed by persistent storage and accessible via system calls.

Tiny ML Embedded Keyword Spotting

Jan 2025 - May 2025

Nanyang Technological University

Singapore

- Deployed real-time audio classification on an Arduino using Keras and TensorFlow for Microcontrollers.
- Designed a quantized CNN with MFCC preprocessing for keyword spotting within 256KB RAM, 78% accuracy.

PUBLICATIONS

Olatunji, S., Tsai, Y.-L., **Gowrishankar, S.**, Bayles, M., Rogers, W., Robots for Older Adults: A Scoping Review. ACM Transactions on Human-Robot Interaction, 2026. <https://doi.org/10.1145/3799977>

LEADERSHIP

President and Exec Advisor

Jan 2024 - Present

Engineers Without Borders at UIUC

Champaign, IL

- Led a board of 22 chapter leaders, overseeing 170+ members and 4 international projects with a \$120K budget.
- Pioneered a membership incentive system, improving membership by 81% and member retention by 26% YoY.
- Led a chapter-wide fundraising team, increasing semesterly funds raised from \$300 to \$30,000 within one semester.
- Traveled to Malawi to collaborate with local experts and implement a vehicular bridge for a village of 25,000.

SKILLS

Languages: Python (Pandas, TensorFlow, scikit-learn, etc.), C++, C, CUDA, RISC-V Assembly

Tools: Azure DevOps, Git, Docker, VSCode, Jupyter, Databricks, Arduino, OpenAI API, Power BI, SQL